CLOUD NATIVE

```
PS D:\kitchen-app> kubectl get nodes
NAME             STATUS   ROLES           AGE     VERSION
docker-desktop   Ready    control-plane   5d19h   v1.34.1
PS D:\kitchen-app>
```

```
PS D:\kitchen-app> kubectl apply -f k8s/namespace.yaml
namespace/kitchen-app unchanged
PS D:\kitchen-app> kubectl apply -f k8s/db/statefulset.yaml
service/mongo unchanged
statefulset.apps/mongo configured
PS D:\kitchen-app> kubectl apply -f k8s/api/deployment.yaml
service/api unchanged
deployment.apps/api-deployment unchanged
PS D:\kitchen-app> kubectl apply -f k8s/web-ui/deployment.yaml
configmap/web-ui-config unchanged
deployment.apps/web-ui-deployment unchanged
service/web-ui-service unchanged
PS D:\kitchen-app>
```

```
PS D:\kitchen-app> kubectl port-forward -n kitchen-app svc/api 5050:3000
Forwarding from 127.0.0.1:5050 -> 3000
Forwarding from [::1]:5050 -> 3000
```

```
PS D:\kitchen-app> Invoke-RestMethod -Method Get -Uri "http://localhost:5050/health"

status db
------ --
ok     connected

PS D:\kitchen-app>
```

```
>> -ContentType "application/json"
>> -Body (@{ dish = "parotta" } | ConvertTo-Json)


_id             : 69c037d24fd29a0f92478c08
dish            : parotta
status          : pending
createdAt       : 2026-03-22T18:41:22.208Z
statusUpdatedAt : 2026-03-22T18:41:22.208Z
nextStatus      : preparing
nextStatusAt    : 2026-03-22T18:41:27.208Z



PS C:\Users\Dharshini\Downloads\Kitchen-app-main> Invoke-RestMethod -Method Post -Uri "http://localhost:3000/orders" `
>> -ContentType "application/json" `
>> -Body (@{ dish = "arisu paruppu" } | ConvertTo-Json)


_id             : 69c037ef4fd29a0f92478c09
dish            : arisu paruppu
status          : pending
createdAt       : 2026-03-22T18:41:51.827Z
statusUpdatedAt : 2026-03-22T18:41:51.827Z
nextStatus      : preparing
nextStatusAt    : 2026-03-22T18:41:56.827Z
```

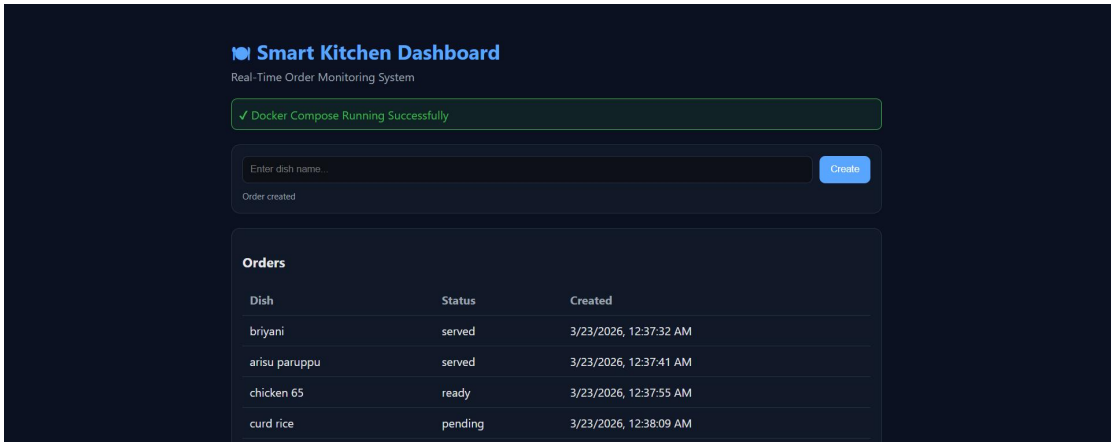
```
PS D:\kitchen-app> kubectl port-forward -n kitchen-app svc/web-ui-service 8080:80
Forwarding from 127.0.0.1:8080 -> 80
Forwarding from [::1]:8080 -> 80
```

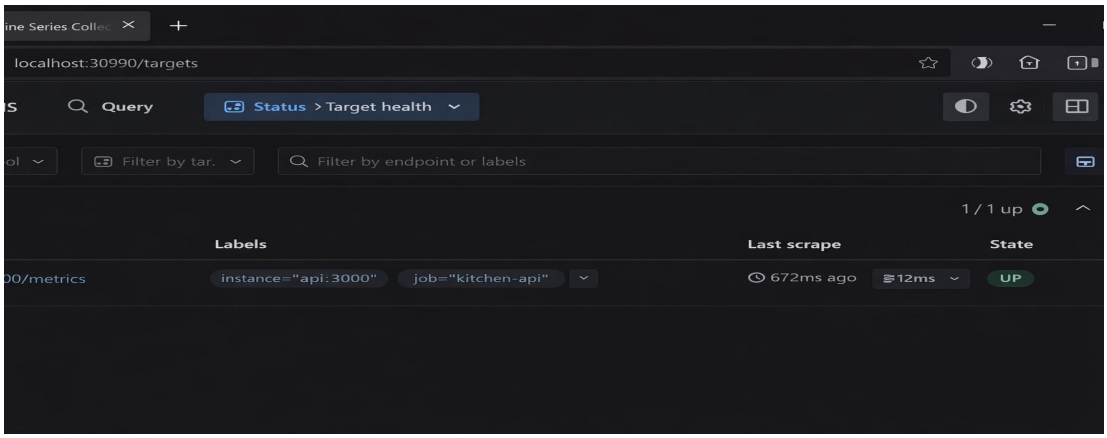
```
PS C:\Users\Dharshini\Downloads\Kitchen-app-main> kubectl get svc -n kitchen-app
NAME                 TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)          AGE
api                  ClusterIP   10.96.55.236    <none>        3000/TCP         8s
grafana-service      NodePort    10.96.240.10    <none>        3000:30300/TCP   41m
mongo                ClusterIP   10.96.220.160   <none>        27017/TCP        9s
prometheus-service   NodePort    10.96.255.183   <none>        9090:30090/TCP   42m
```

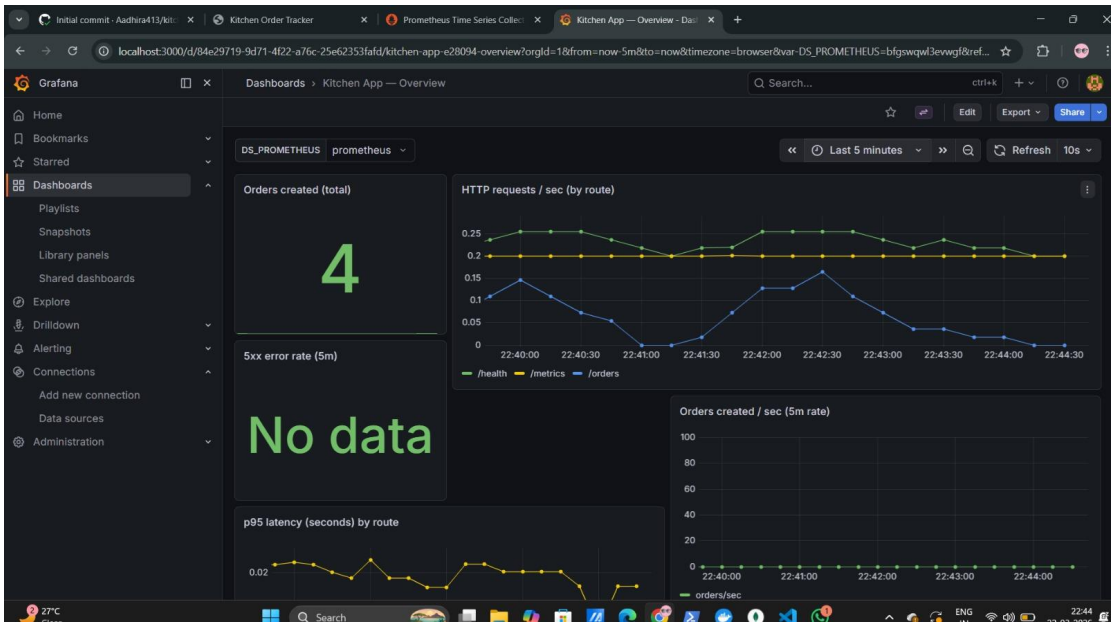
```
PS C:\Users\Dharshini\Downloads\Kitchen-app-main> kubectl port-forward svc/prometheus-service -n kitchen-app 9090:9090
Forwarding from 127.0.0.1:9090 -> 9090
Forwarding from [::1]:9090 -> 9090
Handling connection for 9090
Handling connection for 9090
```

```
PS C:\Users\Dharshini\Downloads\Kitchen-app-main> kubectl port-forward svc/grafana-service -n kitchen-app 3001:3000
Forwarding from 127.0.0.1:3001 -> 3000
Forwarding from [::1]:3001 -> 3000
```

```
PS C:\WINDOWS\system32> cd "C:\Users\Dharshini\Downloads\Kitchen-app-main"
PS C:\Users\Dharshini\Downloads\Kitchen-app-main> kubectl get pods -n kitchen-app
NAME                                  READY   STATUS    RESTARTS   AGE
api-deployment-8687b75877-dzsp8       1/1     Running   0          8m44s
api-deployment-8687b75877-f86bw       1/1     Running   0          8m44s
grafana-deployment-85787b87d4-gdr6m   1/1     Running   0          40m
mongo-0                               1/1     Running   0          8m45s
prometheus-deployment-8478868bbc-g5c8r 1/1    Running   0          7m12s
PS C:\Users\Dharshini\Downloads\Kitchen-app-main>
```


Smart Kitchen Dashboard
Real-Time Order Monitoring System
✓ Docker Compose Running Successfully
Enter dish name...
Create
Order created
Orders
Dish
Status
Created
briyani
served
3/23/2026, 12:37:32 AM
arisu paruppu
served
3/23/2026, 12:37:41 AM
chicken 65
ready
3/23/2026, 12:37:55 AM
curd rice
pending
3/23/2026, 12:38:09 AM


localhost:30990/targets
Query
Status > Target health
Filter by tar.
Filter by endpoint or labels
1 / 1 up
Labels
Last scrape
State
instance="api:3000"
job="kitchen-api"
672ms ago
12ms
UP


Grafana
Dashboards > Kitchen App — Overview
Home
Bookmarks
Starred
Dashboards
Playlists
Snapshots
Library panels
Shared dashboards
Explore
Drilldown
Alerting
Connections
Add new connection
Data sources
Administration
DS_PROMETHEUS
prometheus
Last 5 minutes
Refresh
10s
Edit
Export
Share
Orders created (total)
4
HTTP requests / sec (by route)
/health
/metrics
/orders
5xx error rate (5m)
No data
Orders created / sec (5m rate)
orders/sec
p95 latency (seconds) by route

Grafana
Dashboards › Kitchen App — Overview
Search...
ctrl+k
Home
Bookmarks
Starred
Dashboards
Playlists
Snapshots
Library panels
Shared dashboards
Explore
Drilldown
Alerting
Connections
Add new connection
Data sources
Administration
Edit
Export
DS_PROMETHEUS
prometheus
Last 15 minutes
Refresh
0.02
0.015
0.01
0.005
17:30
17:35
17:40
/health
/metrics
/orders
API RSS memory (bytes)
78000000
76000000
74000000
72000000
70000000
rss
0
orders/sec
API CPU (seconds/sec)
0.018
0.016
0.014
0.012
cpu